

Example Exam with Solutions

Analytical Skills for Business

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Table of contents

Version Control Systems: Git and GitHub	2
Question	2
Programming Languages and Development Environments	2
Question	2
Development environments and tools	3
Question	3
No-Code and Low-Code Tools	3
Question	3
Descriptive Statistics	3
Question	3
Data Quality and Association	4
Question	4
Practical R Programming	4
Part a: Basic Operations	4
Question	4
Part b: Working Directory	4
Question	4
Part c: Package Management	4
Question	5
Part d: Getting Help	5
Question	5

Part e: Data Frame Creation	5
Question	5

References 5

Exam Information

This document contains example exam questions with detailed solutions for the Analytical Skills for Business course at Hochschule Fresenius.

Total Points: 90

Duration: 90 minutes (1 point per minute) **Questions:** 7 sections

Version Control Systems: Git and GitHub

Points: 15

Question

- (5 points) Explain the key differences between Git and GitHub. What is the primary purpose of each?
- (5 points) Describe three core concepts in Git (repository, commit, branch) and explain their importance in collaborative analytics projects.
- (5 points) Explain the difference between merging and rebasing in Git. When would you use one approach over the other?

Programming Languages and Development Environments

Points: 12

Question

- (6 points) Compare R, Python, and SQL in terms of their primary use cases in business analytics. Provide one specific example for each language.
- (6 points) Explain three elements of programming languages (from: syntax, libraries, variables, functions, objects, conditions, loops) and why they are important for data analysis.

Development environments and tools

Points: 15

Question

- a) (7 points) Define what containers like Docker do in the context of software development and data analytics. Explain their benefits.
- b) (8 points) Explain what an IDE is and name two popular IDEs used for data analytics. Describe one key feature of each.

No-Code and Low-Code Tools

Points: 10

Question

- a) (5 points) Name and briefly describe three no-code or low-code tools for data analytics covered in the course (from: n8n, Snowflake, QGIS, Tableau, Power BI, KNIME).
- b) (5 points) Explain the advantages of using no-code/low-code tools in business analytics contexts and when they might be preferred over traditional programming approaches.

Descriptive Statistics

Points: 18

Question

- a) (9 points) Explain three measures of central tendency (mean, median, mode) and when each is most appropriate to use.
- b) (9 points) Distinguish between univariate, bivariate, and multivariate data analysis. Provide one business example for each.

Data Quality and Association

Points: 12

Question

- a) (6 points) Describe three common data quality issues (from: missing values, inconsistent data, duplicate records, outliers) and appropriate strategies for handling each.
- b) (6 points) Explain the difference between correlation and regression analysis. Why is it important to remember that “correlation does not imply causation”?

Practical R Programming

Points: 8

Part a: Basic Operations

Points: 2

Question

Write R code to calculate and store the following result in a variable named `result`:

$$3 + \sqrt{9} - 2^2 \cdot \frac{4}{3}$$

Part b: Working Directory

Points: 1

Question

Write the R code to get the path of your current working directory.

Part c: Package Management

Points: 1

Question

Write the R code to install the `tidyverse` package.

Part d: Getting Help

Points: 1

Question

Write the R code to access the help documentation for the `mean()` function.

Part e: Data Frame Creation

Points: 3

Question

Create a data frame in R containing the following sales data:

Region	Sales
North	45000
South	38000
East	52000

Store it in a variable named `sales_data`.

References

Cetinkaya-Rundel, M., & Hardin, J. (2021). *Introduction to modern statistics* (2nd ed.). OpenIntro.

Illowsky, B., & Dean, S. (2018). *Introductory statistics*. OpenStax.

GeeksforGeeks. (2024). *Understanding hypothesis testing*. <https://www.geeksforgeeks.org/software-testing/understanding-hypothesis-testing/>

GitHub Docs. (2024). *About GitHub and Git*. <https://docs.github.com/en/get-started/start-your-journey/about-github-and-git>